

Branch: Computer Science and Engineering
Course Title: Data Structures **Course Code:** PCC CS-301
Total Marks: 200 **Internal Marks:** Theory-50 Practical-50

Class: B.Tech. 3rd Sem.
Credits: 4 **L-T-P:** 3-0-2
External Marks: 100

Course objectives:

- To understand the basic concepts of data structures and algorithms.
- To differentiate linear and non-linear data structures and the operations upon them.
- Ability to perform sorting and searching in a given set of data items.
- To comprehend the necessity of time complexity in algorithms.

Course Outcome:

- Understanding the fundamental analysis and time complexity for a given problem.
- Understanding concept of linear & non-linear data structures and legal operations permitted on them.
- Applying a suitable algorithm for searching.
- Applying a suitable algorithm for sorting.
- Understanding the concept of graph data structures, such as trees, binary trees, graphs and AVL trees and operations on them.

Unit- I: Introduction to Data Structures - Introduction: Structured programming concepts, Basic concept of data structures and pointers. Dynamic memory management in C. Fundamentals of algorithm analysis, Space and time complexity of an algorithm, Types of asymptotic notations and orders of growth, Algorithm efficiency – best case, worst case and average case.

Abstract Data-types: Arrays, 1D and 2D array, Stack - Applications of stack: Expression Evaluation - Conversion of Infix to postfix and prefix expression, Tower of Hanoi.

Queue - Types of Queues: Circular Queue, Double Ended Queue (deQueue), Applications – Priority Queue using Arrays

Unit -II: Linear Data Structures and Files - Linked List and its comparison with array implementation. Types of Linked lists, Applications of Linked lists. Implementing Linked Lists using structures. Insertion, Deletion, Search, display and reverse in single and doubly linked lists.

Linked List implementation of Stack, Queues, Circular Queues. Polynomial manipulation using Linked lists.

Unit -III: Sorting and Searching- Searching - Linear Search and binary search, Applications - Finding square root of 'n' - Longest Common Prefix

Sorting – Insertion sort - Selection sort – Bubble sort – (Counting Sort) - Quick sort- Merge sort, Shell Sort, radix sort, Heap sort algorithms and their Analysis.

File Organization - Organization (Sequential, Direct, Indexed Sequential, Hashed) and various types of accessing schemes.

Unit-IV:Non- linear Data Structures - Graph Algorithms: Graphs and their Representations- Adjacency list and adjacency matrix representations. Basic graph definitions: paths, cycles, trees, spanning trees, connected components, Euler's formula. Graph Traversal Techniques: Breadth First Search (BFS) and Depth First Search (DFS), Applications of BFS and DFS, Minimum Spanning Trees (MST), Prim's and Kruskal's algorithms for MST, Connected Components, Dijkstra's Algorithm for Single Source Shortest Paths.

Trees: Terminology, Binary Tree – Terminology and Properties, Tree Traversals, Expression Trees – Binary Search Trees – operations in BST – insertion, deletion, finding min and max. AVL trees and rotations in AVL trees.

Hashing - Hash functions, open hashing and closed hashing techniques.

Books Recommended:

Text Books

1. E Horowitz and S Sahni, “Fundamentals of Data Structures”, Second Edition, Galgotia Books, 2008.
2. Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman, “Data Structures and Algorithms”, First Edition, Pearson Publishers, 1983.

References:

1. Knuth Donald E, “Art of Computer Programming: Fundamental Algorithms Volume 1
2. Neapolitan, R., & Naimipour, K. (2010). Foundations of algorithms. Jones & Bartlett Publishers.
3. Pat Morin, Open Data Structures: An Introduction (Open Paths to Enriched Learning), 31st ed. Edition, UBC Press, 2013.
4. Data Structures with C, Schaum's Outline Series.

Web References:

<https://nptel.ac.in/courses/106102064>

<https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/>

https://onlinecourses.swayam2.ac.in/cec19_cs04/preview

https://web.iitd.ac.in/~csz188551/COL106_2019/

List of Practicals:

The practicals are based on the topics covered under Theory Syllabus. The students are required to perform following Programs.

1. Write a program to check if expression is correctly parenthesized using Stacks.
2. Write a program to evaluate Postfix Expression using Stacks.
3. Write a program to convert Infix Expression to its corresponding Postfix and Prefix.
4. Write a program to convert Prefix Expression to Postfix.
5. Write a program to implement Circular Queue Operations.
6. Write a program to implement Priority Queue Operations.
7. Write a program to implement operations on Doubly Linked List.
8. Write a program to Sort a list of numbers using Binary Search Tree.
9. Write a program to implement QuickSort algorithm.
10. Write a program to implement Shell Sort algorithm.
11. Write a program to implement Merge Sort algorithm.
12. Implement Binary search trees and their traversals in C.
13. Implement Graphs and Graph Traversals (BFS, DFS).